Roll No. _____

28 MAR 2019

B.C.A Sem-3 Examination

Subject Code: 20630

Operating System

Total: 70 Marks **Time: 2 ½ Hrs**

Q-1 [a] What is operating system? Explain its all function. 07

Q-1 [b] Write a sort note on real time system. 07

OR

Q-1 [a] What is virtual machine & client server? 07

Q-1 [b] What is monolithic layered and jobs? 07

Q-2 [a] What is process life cycle? 07

Q-2 [b] Write a sort note on CPU utilization & response time. 07

OR

Q-2 [a] Explain process control block. 07

Q-2 [b] Explain FCFS Algorithm with Following Example: 07

Find Out Waiting Time, Average Waiting Time, Turnaround Time, and Average Turnaround Time (Time Quantum Is 3ms)

Process	CPU Burst Time	Arrival Time
P1	8	5
P2	5	0
P3	4	4
P4	6	8
P5	4	3

Q-3 [a] What is thread? Explain its types. 07

Q-3 [b] State the difference between process & thread. 07

OR

Q-3[a] Explain deadlock recovery & avoidance. 07

Q-3[b] Difference between user level & kernel level thread. 07

Q-4[a] What is fragmentation? 07

Q-4[b] Explain logical & physical address map. 07

OR

Q-4[a] Explain internal & external fragmentation. 07

Q-4[b] What is paging? Explain its disadvantages. 07

Q-5[a] Explain Segmentation 07

Q-5[b] Explain Virtual memory. 07

OR

Q-5[a] Explain page replacement policies. 07

Q-5[b] How to memory protection in paging? 07